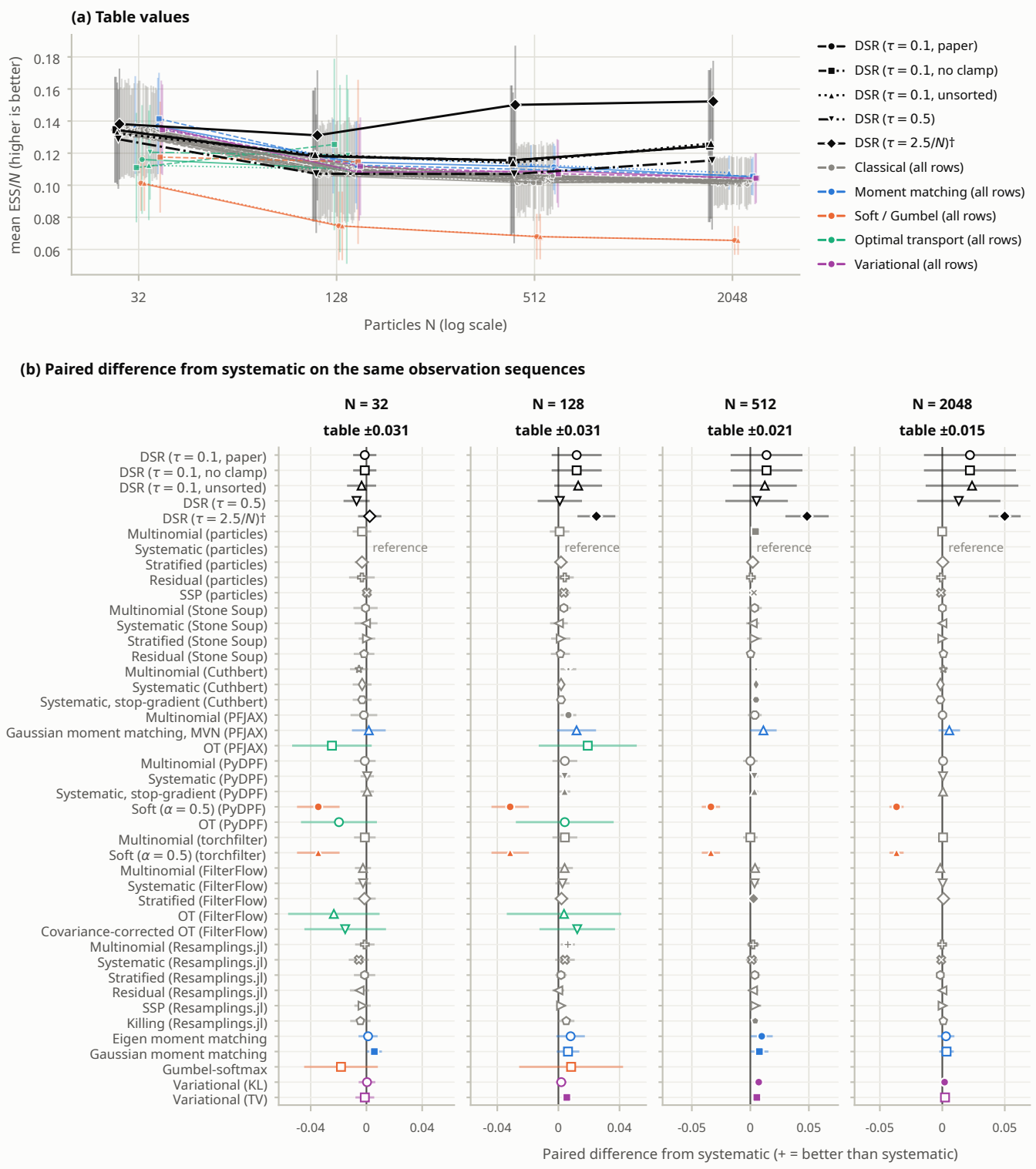

(a) Table values
mean ESS/N (higher is better)
0.18
0.16
0.14
0.12
0.10
0.08
0.06
32
128
512
2048
Particles N (log scale)
DSR (τ = 0.1, paper)
DSR (τ = 0.1, no clamp)
DSR (τ = 0.1, unsorted)
DSR (τ = 0.5)
DSR (τ = 2.5/N)†
Classical (all rows)
Moment matching (all rows)
Soft / Gumbel (all rows)
Optimal transport (all rows)
Variational (all rows)
(b) Paired difference from systematic on the same observation sequences
N = 32
table ±0.031
N = 128
table ±0.031
N = 512
table ±0.021
N = 2048
table ±0.015
reference
DSR (τ = 0.1, paper)
DSR (τ = 0.1, no clamp)
DSR (τ = 0.1, unsorted)
DSR (τ = 0.5)
DSR (τ = 2.5/N)†
Multinomial (particles)
Systematic (particles)
Stratified (particles)
Residual (particles)
SSP (particles)
Multinomial (Stone Soup)
Systematic (Stone Soup)
Stratified (Stone Soup)
Residual (Stone Soup)
Multinomial (Cuthbert)
Systematic (Cuthbert)
Systematic, stop-gradient (Cuthbert)
Multinomial (PFJAX)
Gaussian moment matching, MVN (PFJAX)
OT (PFJAX)
Multinomial (PyDPF)
Systematic (PyDPF)
Systematic, stop-gradient (PyDPF)
Soft (α = 0.5) (PyDPF)
OT (PyDPF)
Multinomial (torchfilter)
Soft (α = 0.5) (torchfilter)
Multinomial (FilterFlow)
Systematic (FilterFlow)
Stratified (FilterFlow)
OT (FilterFlow)
Covariance-corrected OT (FilterFlow)
Multinomial (Resamplings.jl)
Systematic (Resamplings.jl)
Stratified (Resamplings.jl)
Residual (Resamplings.jl)
SSP (Resamplings.jl)
Killing (Resamplings.jl)
Eigen moment matching
Gaussian moment matching
Gumbel-softmax
Variational (KL)
Variational (TV)
-0.04
0
0.04
-0.05
0.05
Paired difference from systematic (+ = better than systematic)